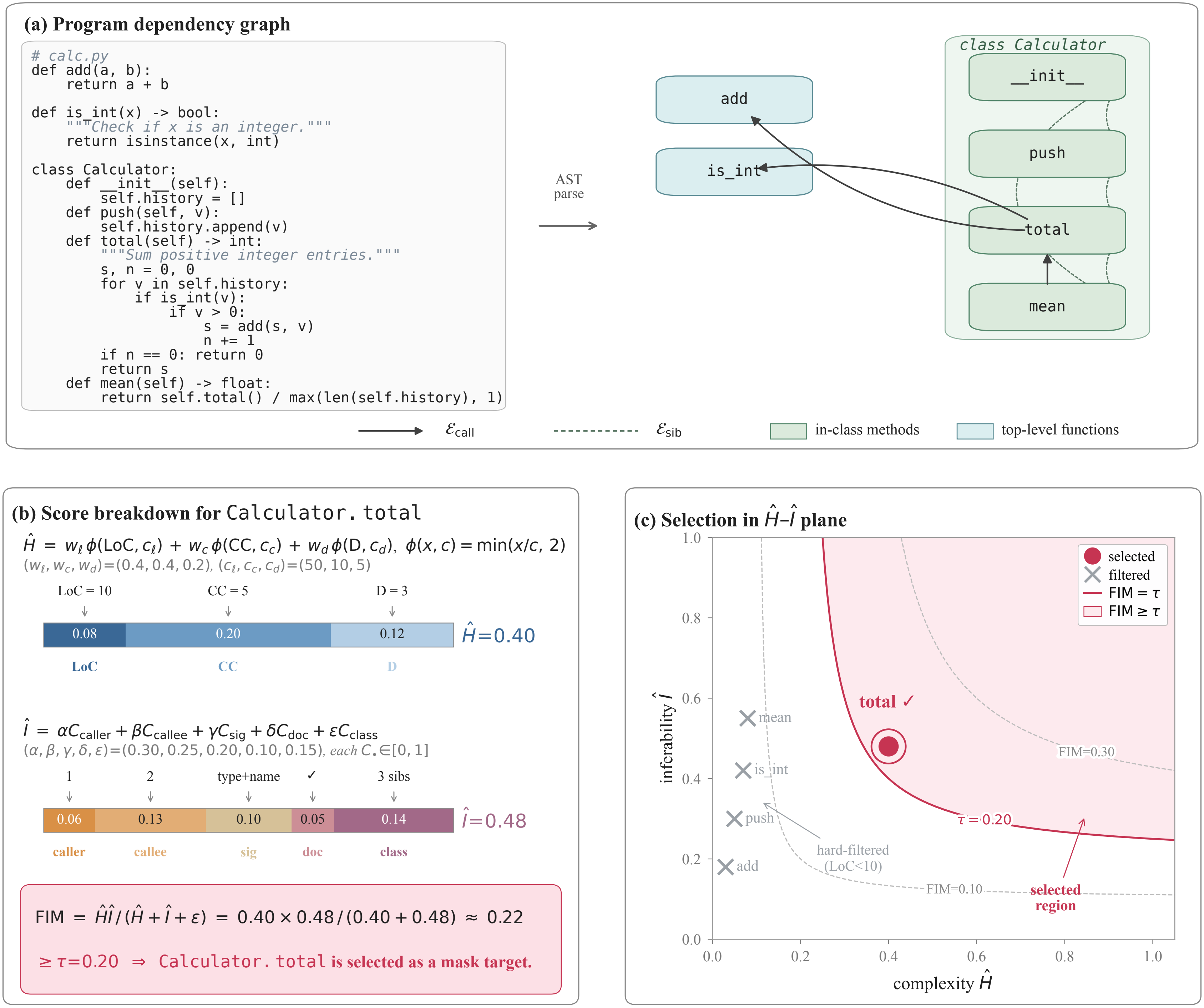

(a) Program dependency graph
# calc.py
def add(a, b):
    return a + b
def is_int(x) -> bool:
    """Check if x is an integer."""
    return isinstance(x, int)
class Calculator:
    def __init__(self):
        self.history = []
    def push(self, v):
        self.history.append(v)
    def total(self) -> int:
        """Sum positive integer entries."""
        s, n = 0, 0
        for v in self.history:
            if is_int(v):
                if v > 0:
                    s = add(s, v)
                    n += 1
        if n == 0: return 0
        return s
    def mean(self) -> float:
        return self.total() / max(len(self.history), 1)
AST parse
add
is_int
class Calculator
__init__
push
total
mean
$\mathcal{E}_{\text{call}}$
$\mathcal{E}_{\text{sib}}$
in-class methods
top-level functions
(b) Score breakdown for Calculator.total
$\hat{H} = w_\ell\,\phi(\mathrm{LoC}, c_\ell) + w_c\,\phi(\mathrm{CC}, c_c) + w_d\,\phi(\mathrm{D}, c_d),\ \phi(x, c) = \min(x/c, 2)$
$(w_\ell, w_c, w_d) = (0.4, 0.4, 0.2),\ (c_\ell, c_c, c_d) = (50, 10, 5)$
LoC = 10
CC = 5
D = 3
0.08
0.20
0.12
$\hat{H}$=0.40
LoC
CC
D
$\hat{I} = \alpha C_{\text{caller}} + \beta C_{\text{callee}} + \gamma C_{\text{sig}} + \delta C_{\text{doc}} + \varepsilon C_{\text{class}}$
$(\alpha, \beta, \gamma, \delta, \varepsilon) = (0.30, 0.25, 0.20, 0.10, 0.15)$, each $C_\bullet \in [0, 1]$
1
2
type+name
✓
3 sibs
0.06
0.13
0.10
0.05
0.14
$\hat{I}$=0.48
caller
callee
sig
doc
class
$\mathrm{FIM} = \hat{H}\hat{I} / (\hat{H} + \hat{I} + \varepsilon) = 0.40 \times 0.48 / (0.40 + 0.48) \approx 0.22$
$\geq \tau = 0.20 \Rightarrow$ Calculator.total is selected as a mask target.
(c) Selection in $\hat{H}$–$\hat{I}$ plane
selected
filtered
FIM = τ
FIM ≥ τ
total ✓
mean
is_int
push
add
FIM=0.30
τ = 0.20
FIM=0.10
hard-filtered (LoC<10)
selected region
inferability $\hat{I}$
complexity $\hat{H}$
0.0
0.2
0.4
0.6
0.8
1.0